

MASTER OF COMPUTER APPLICATIONS (2024 – 2026)

Project Title: *Academic Collaboration Platform Using MERN Stack*

Organization: ConsultIT

Guided By:

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Submitted By:

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Problem Statement

Academic research collaboration in many institutions often relies on informal communication and manual coordination between students and professors. This leads to missed opportunities, inefficient project allocation, and poor tracking of research progress. A centralized digital platform is required to streamline collaboration and research management.

Objectives

- To develop a secure web-based academic collaboration platform.
- To implement role-based access for Students, Professors, and Administrators.
- To recommend research projects to students based on their skills and interests.
- To provide real-time collaboration tools for research teams.

Scope of the Project

The platform enables professors to post research projects and allows students to explore and apply for them. It also provides a collaborative workspace with features such as project applications, approval workflow, real-time chat, task tracking, and discussion forums. The system is designed to support future enhancements in academic research management.

Role in the Project: Mern Stack Developer

As a Mern Stack Developer, I am responsible for the end-to-end development of the application. This includes designing and implementing the frontend user interface, developing backend APIs, creating and managing the database schema, implementing authentication and authorization mechanisms, integrating real-time collaboration features, and deploying the application in a live production environment.

Project Modules:

Module 1: Authentication & Role-Based Access Control (RBAC)

Module 2: Project Management & AI Matching Engine

Module 3: Application Workflow & Admin Oversight

Module 4: Real-Time Collaboration Workspace

Milestone Done!

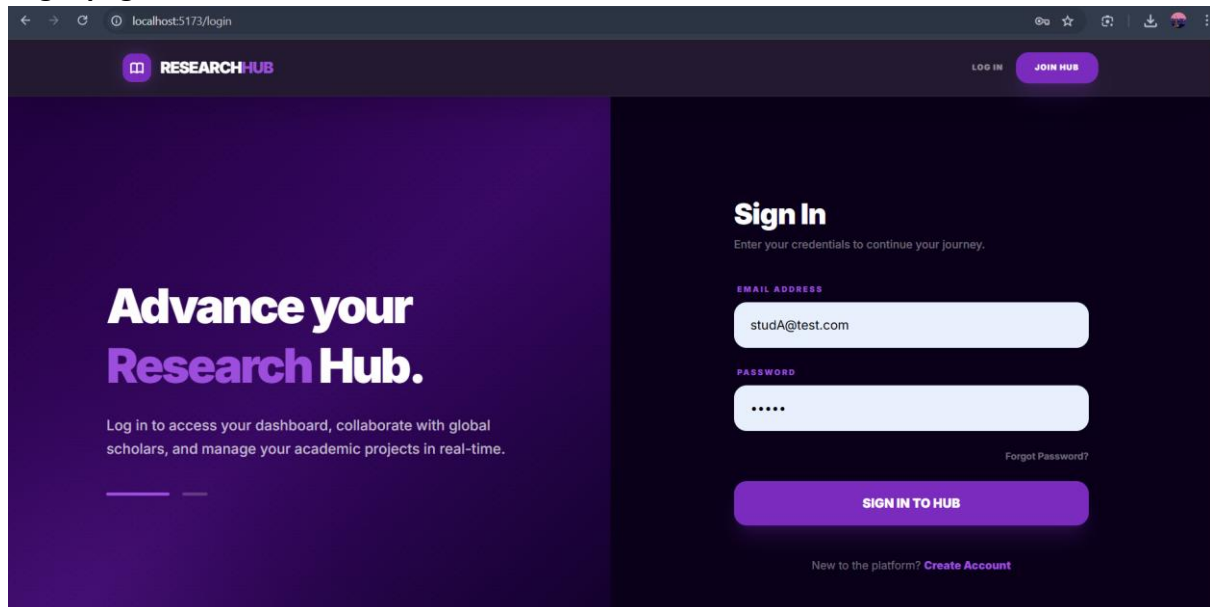
Module 1, 2 and 3 has been done:

1. Authentication & Role-Based Access Control (RBAC)
2. Project Management & AI Matching Engine
3. Real-Time Collaboration Workspace

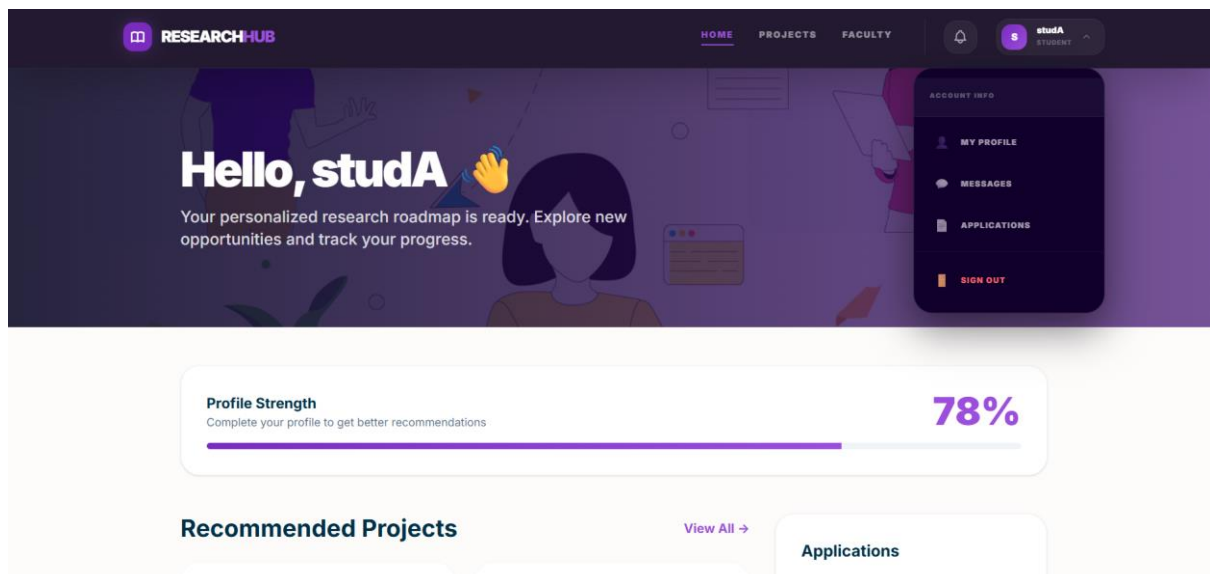
Status: COMPLETE

SOME SNAPSHOTS

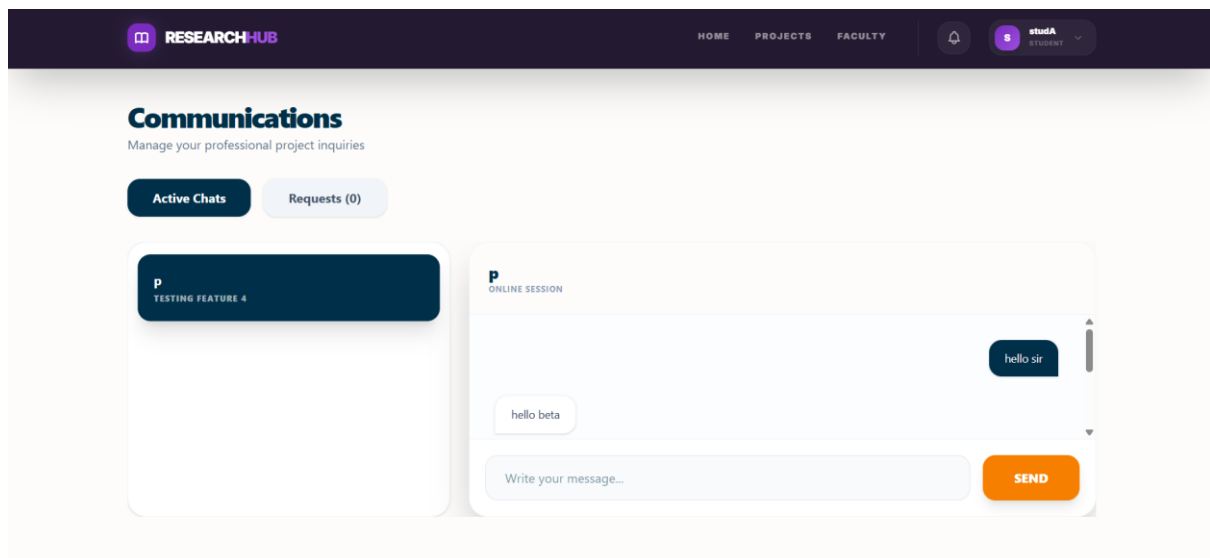
Login page



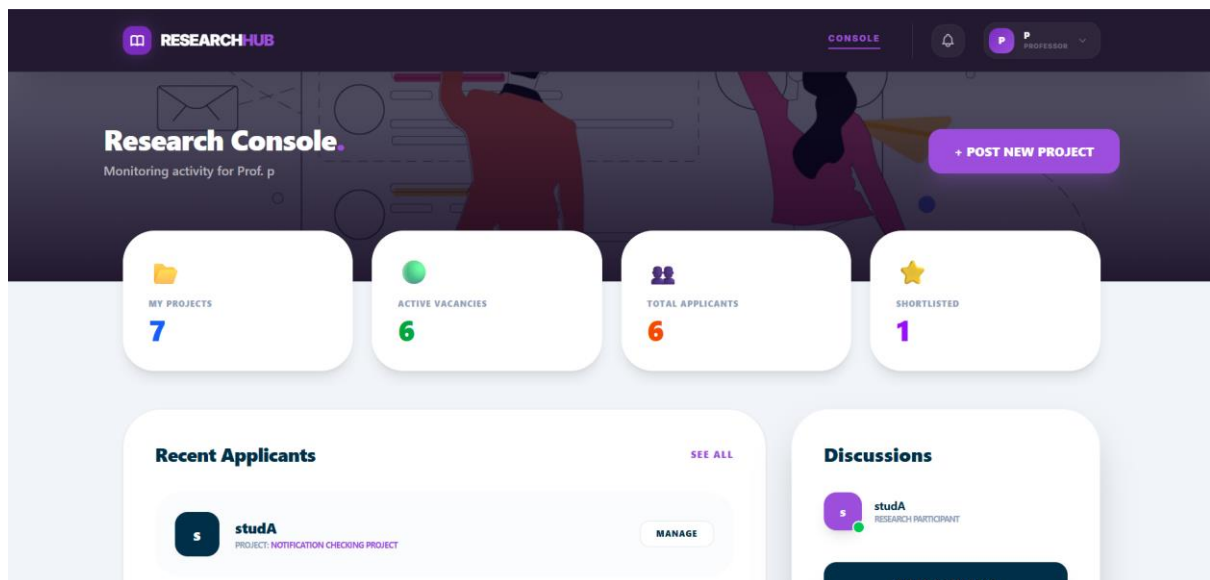
Student dashboard



Chat section



Professor dashboard



Create new project

RESEARCHHUB

CONSOLE

P

PROFESSOR

Post New Research Project

Project Title

Duration

3 months

Description

Required Skills (comma separated)

Python, Machine Learning, NLP

Research Field

AI, Computer Vision

Mode

Remote

Paid / Unpaid

Unpaid

Deadline

dd-mm-yyyy

Post Project

Matching score

RESEARCHHUB

HOMEPROJECTSFACULTY

S

studA

STUDENT

Recommended Projects

react project

qwertyulop

100% MATCH SCORE

Web Development Platform

second need

71% MATCH SCORE

testing feature 4

ww

71% MATCH SCORE

new

new

0% MATCH SCORE

[View All →](#)

Recommended Faculty

p

cdac

0% Match

Dr. Priya Sharma

0% Match

Dr. Amit Ve

0% Match

[View All →](#)

Applications

0

APPLIED

0

ACCEPTED

[VIEW ALL APPLICATIONS](#)

Saved Projects

No saved projects yet

Recent Activity

- You applied to "AI Research Assistant"
- Professor Sharma viewed your profile
- New message from Dr. Priya Patel

Code base

